

AWS Cloud Data Engineering Project Portfolio

Showcasing 5 real-world data pipelines built using AWS managed services for automation, analytics, and scalability.



1. Insurance Claims Processing & Analytics Pipeline

Domain: Insurance | **Type:** Event-Driven + Batch

Overview:

Built a serverless claims processing system that validates, classifies, and routes insurance claims using AWS Lambda, RDS, and SQS. Real-time dashboards in QuickSight track claim approvals, payment breakdowns, and insurer performance.

AWS Services: S3, Lambda, RDS, SQS, DynamoDB, Glue, QuickSight, KMS

Outcome: Automated claim validation, reduced manual errors, improved traceability, and achieved real-time analytics.



2. Energy Consumption Metering (IoT Time-Series Pipeline)

Domain: IoT / Energy | **Type:** Streaming + Batch

Overview:

Developed a scalable system to monitor household and industrial energy usage. Implemented real-time data ingestion with Kinesis, nightly aggregation using Glue, and automated anomaly detection with SNS alerts.

AWS Services: Kinesis, Firehose, DynamoDB, Glue, API Gateway, SNS, Athena

Outcome: Enabled real-time usage monitoring, billing automation, and anomaly alerts with optimized data storage.



3. Financial Trades Settlement Pipeline

Domain: Finance | **Type:** Batch ELT (Serverless)

Overview:

Automated daily trade settlements and P&L calculations using Redshift and Athena. Transformed broker trade data from S3 into structured insights with QuickSight dashboards for profitability and trade trends.

AWS Services: S3, Glue, Athena, Redshift Serverless, QuickSight

Outcome: Reduced settlement time, improved compliance, and provided real-time trade visibility.



4. Automated Retail Inventory Reconciliation & Alert System

Domain: Retail | **Type:** Batch + Event-Driven

Overview:

Designed an automated system to reconcile daily retail sales and stock data. Used Glue ETL for reconciliation, Athena for analytics, and SNS for alerts on oversold inventory.

AWS Services: S3, Glue, Athena, Lambda, SNS, QuickSight

Outcome: Eliminated manual reconciliation, improved operational accuracy, and provided store-level BI insights.



5. Food Delivery Order Pipeline

Domain: FoodTech | **Type:** Real-time Streaming

Overview:

Developed a real-time data pipeline to capture and analyze food order lifecycles. Leveraged DynamoDB Streams, Kinesis Firehose, and Glue for transformation with Athena and QuickSight for analytics.

AWS Services: DynamoDB, Streams, Lambda, Firehose, S3, Glue, Athena, QuickSight

Outcome: Delivered real-time order tracking, performance monitoring, and restaurant efficiency insights.



Summary Table

Project	Domain	Type	Core AWS Services	Key Outcome
Insurance Claims	Insurance	Serverless + Batch	Lambda, RDS, QuickSight	Automated claims pipeline
Energy Metering	IoT/ Energy	Streaming + Batch	Kinesis, Glue, SNS	Real-time billing & anomaly alerts
Financial Trades	Finance	Batch ELT	S3, Athena, Redshift	Automated P&L computation

Project	Domain	Type	Core AWS Services	Key Outcome
Retail Inventory	Retail	Event-driven	Glue, SNS, Athena	Inventory reconciliation automation
Food Delivery	FoodTech	Real-time	DynamoDB, Firehose, Glue	Real-time delivery analytics

Key Skills

AWS Glue | Athena | Redshift | Kinesis | DynamoDB | Lambda | QuickSight | S3 | EventBridge | Data Lake Design | ETL Automation | Real-time Analytics

About Me

I am passionate about building scalable, automated cloud data solutions using AWS. I focus on real-time analytics, data orchestration, and cloud cost optimization — bridging the gap between data engineering and business insights.

 **Let's connect and collaborate on cloud data engineering opportunities!**